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Identification of nutrient deficiency stress for iron, zinc and manganese in baby spinach using computer vision

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ABSTRACT

In recent years, various methodologies have been developed to evaluate the quality of agricultural products. These technologies have facilitated the monitoring of plant nutritional needs, thereby reducing reliance on human judgment and enhancing intuitive decision-making. This study focuses on the cultivation of baby spinach plants in soilless beds to precisely control nutrient conditions. The objective was to investigate deficiencies for three key micronutrients, namely, iron (Fe), zinc (Zn), and manganese (Mn), at five concentration levels, i.e., 0 %, 25 %, 50 %, 75 %, and 100 % of the recommended amount. A set of color, texture and morphological features were extracted from the sample images. Feature selection was carried out using the ReliefF method and the Sequential selection method for four distinct modes: Sequential Forward Selection (SFS), Sequential Backward Selection (SBS), Sequential Floating Forward Selection (SFFS) and Sequential Floating Backward Selection (SFBS). Results showed that SFBS exhibited the best performance compared to the other algorithms. Subsequently, five classification methods (Artificial Neural Network (ANN), Naive Bayes (NB), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest (RF)) were evaluated using the selected features as input in a predetermined order based on the SFBS results. The best rankings were ANN, KNN, RF, SVM and NB, respectively. Hence, as soon as the true leaf of a baby spinach crop expands, the deficiency of three micronutrients can be detected by ANN at various levels with an accuracy of 88 % for Fe, 84 % for Mn and 76 % for Zn. A hierarchical model was also applied, achieving over 70 % accuracy.

1. Introduction

For the production of agricultural commodities, especially fruits and vegetables, it is necessary to adopt new cultivation methods in greenhouse environments, in order to avoid the low-yield and traditional methods of outdoor cultivation. Greenhouse cultivation, despite the high investment costs and the need for continuous monitoring, increases productivity, reduces damage caused by cold, heat, rainfall and other unexpected weather events, saves water and energy and can bring the product to the market early. Furthermore, the properties of different materials in the cultivation bed have a direct and indirect effect on the growth and production of the crop. Therefore, combining greenhouse cultivation with new technologies such as soilless cultivation (hydroponics) has provided the possibility of better control of environmental conditions, moisture and plant nutrition during the growing season [1,2]. However, it is also essential to determine the different needs of plants during the growth process, and to take the necessary actions to meet the specific quantitative and qualitative needs of the plant to prevent quality loss. Early detection of nutrient deficiency in plants is one of the most important issues; if this stress is ignored growth and product quality can be seriously impacted [3]. Measuring plant nutrients through destructive methods in the laboratory is usually time-consuming and expensive. Therefore, a non-destructive and low-cost method is needed to carefully monitor plant conditions [4]. Based on the advantages of plant monitoring systems, commercialization of intelligent systems for greenhouse production has evolved. The

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high demand for producing more crops per unit area, and the need to save energy and water and to reduce production costs, has made the use of new technologies in agriculture inevitable.

Baby spinach (L. *Chenopodiacea Spinacea oleracea*) is a type of leafy vegetable that is rich in several vitamins including A, B1, B2 and C and, mineral elements such as iron, calcium and magnesium. This plant has a crunchy texture and tender leaves, and its consumption has spread rapidly throughout the world due to its positive effects on human health [5,6]. Insufficient nutrients is one of the main causes for crop damage. In general, a group of specific elements are needed by all plants during the growth process. The required amount of these elements varies for each species, and they are divided into two categories: macro elements, i.e., nitrogen, phosphorus, potassium, calcium, magnesium and sulfur, and micro elements, i.e., iron, manganese, zinc, boron, copper, molybdenum and chlorine [3]. High amounts of nitrogen fertilizers are usually used in the production of leafy vegetables, especially spinach. This excessive use of nitrogen fertilizers can cause environmental problems through leaching and greenhouse gas emissions, and decreases the amount of some nutrients such as Fe and Mn in spinach [7,8]. Additionally, factors such as unbalanced soil nutrient levels, high soil pH and calcareous soil, excessive irrigation and soil erosion, low soil temperature, and poor air circulation contribute to nutrient deficiencies, particularly for Fe, Mn and Zn [9,10].

Most plants require Fe and it is one of the elements that plays an important role in the production of chlorophyll that affects photosynthesis and respiration. Fe deficiency is first observed in young leaves and in the upper parts of the stem, and gradually affects the entire plant with an increase in the deficiency [11]. Plants also require Zn to form plant protein and regulate the growth process. The application of a sufficient amount of Zn increases the amount of protein and fiber in spinach [12]. Senescence of young leaves, growth termination, and thinning and brittleness of stems are common symptoms for Zn deficiency [13]. Mn is another essential element for plant growth, playing a crucial role in seed formation, enzyme processes, protein metabolism and chlorophyll production. It enhances germination and aids in water decomposition reactions that produce oxygen. The most noticeable symptom of Mn deficiency is interveinal chlorosis typically seen in the upper parts of the plant. Although symptoms of Fe and Mn deficiencies are similar, Mn deficiency can be distinguished by the green tips and margins of the leaves [9,14].

Analyzing morphological characteristics is crucial for detecting various nutrient deficiencies in plants [15]. Recent research has focused on using color, morphological and texture features to detect stress in plants [16–20]. Several studies have explored the use of various techniques to detect nutrient deficiencies in plants. For instance, color, texture, and shape parameters have been used to identify phosphorus, potassium, and nitrogen deficiencies in rice plants with an accuracy of 87.5 % using the learning vector quantization (LVQ) method [21]. Deep convolutional neural networks have also been applied to monitor the nutrient status in lettuce, achieving a 96.5 % accuracy in distinguishing different nutrient deficiencies [22]. Guerrero et al [23], used convolutional neural networks trained with transfer learning to identify nutrient deficiencies in banana leaves with a 100 % accuracy. Tuan et al [24] developed an intelligent tool based on deep kernel learning to predict magnesium, phosphate, and calcium concentrations in lettuce, showing optimal performance with RMSE values of 12.5, 12.1, and 7.5 mg/liter, respectively, and a coefficient of variation below 10 %. Story et al [25] utilized machine vision and regression analysis to detect calcium deficiency in lettuce with an R^2 of 97 %. Fuentes-Pacheco et al [26] examined nitrogen concentration in basil plants using deep learning, achieving an average detection accuracy 7 % higher than traditional methods. Ramdan et al [27] used neural network algorithms to recognize Fe deficiency in Pak Choy, reaching a detection accuracy of 98 % with the MobileNetV2 architecture. Espejo-Garcia et al [28] used RGB imaging to identify nutrient deficiencies in sugar beet and orange trees with accuracies of 98.65 % and 98.52 %, respectively. In another study, boron deficiency in corn was identified using principal component analysis and KNN methods, achieving a 98 % detection accuracy [29].

The goal of this research is to design and develop a non-destructive sensing system for greenhouse production of baby spinach to detecting nutrient deficiencies, specifically Fe, Mn, and Zn. This system will enable close monitoring of crop growth, facilitating timely interventions to meet the plants' needs. As a result, vegetable productivity and produce quality for greenhouse production systems can be significantly improved.

2. Materials and methods

The stages for detecting nutrient deficiencies, i.e., Fe, Zn, and Mn, in baby spinach are illustrated in Fig. 1. Images of the samples were captured under consistent lighting conditions, from which a set of color, morphological, and texture features were extracted. After evaluating some feature selection algorithms (ReliefF, SFS, SBS, SFFS, and SFBS), the features were sorted in order of superiority, and used as input for several classifiers, including ANN, KNN, NB, SVM, and RF. Subsequently, by considering different output classes of nutrient deficiencies in baby spinach plant, the performance of these classifiers were compared. The best classification method was ultimately selected for the timely detection of Fe, Zn, and Mn deficiencies in baby spinach throughout the cultivation period. Each of these steps is detailed in the following sections.

2.1. Experimental design

To study the independent effects of Fe, Zn and Mn nutrients on baby spinach, five nutrient levels were used, i.e. 0 %, 25 %, 50 %, 75 % and 100 % of the recommended amount. The treatments consisted of a control group (100 % recommended level), where plants received the recommended levels of all nutrients, and four deficiency stress levels for each of the nutrients Fe, Zn, and Mn (Fe 0 %, Fe 25 %, Fe 50 %, Fe 75 %; Mn 0 %, Mn 25 %, Mn 50 %, Mn 75 %; and Zn 0 %, Zn 25 %, Zn 50 %, Zn 75 %). With six replications for each treatment, a total of 78 baby spinach samples were cultivated under hydroponic conditions in the greenhouse. The temperature in the greenhouse was maintained between 10 and 20 $^{\circ}$C and a relative humidity of 55±5 % over a period of six weeks (Fig. 2). Imaging was initiated when the first true leaf of baby spinach was fully developed, which is defined as Day 1 of the cultivation period. It is noteworthy that a total of 1092 images were analyzed in this study.

The plants were grown in styrofoam seedling trays containing perlite, with a dimension of 57 × 32 cm. Following the seed germination stage, these trays were placed on plastic boxes containing a nutrient solution. The plastic boxes, made of polyethylene, had a dimension of 60 × 40 × 31 cm. The nutrient solution was prepared using Hoagland's formula, a widely used solution for growing vegetables in a hydroponic system [30]. In the control group, all nutrients were supplied at the recommended levels of the Hoagland solution. Deficiency treatments were applied by altering only the concentration of the nutrient under investigation (Fe, Zn, or Mn) to 0, 25, 50, and 75 % of the recommended level, while the concentrations of all other nutrients were maintained at recommended levels. In this way, the effects of each nutrient were assessed independently. The experiment was in the research greenhouse of the Agricultural Sciences and Natural Resources University of Khuzestan, Iran.

2.2. Image acquisition and analysis

Images were captured under constant lighting conditions in the greenhouse. To ensure consistent illumination, all images were taken at 8:00 AM throughout the cultivation period. Ambient light intensity at this time was measured using a light meter, and was approximately 3365 lx. Imaging was performed every two days over the 6-week cultivation

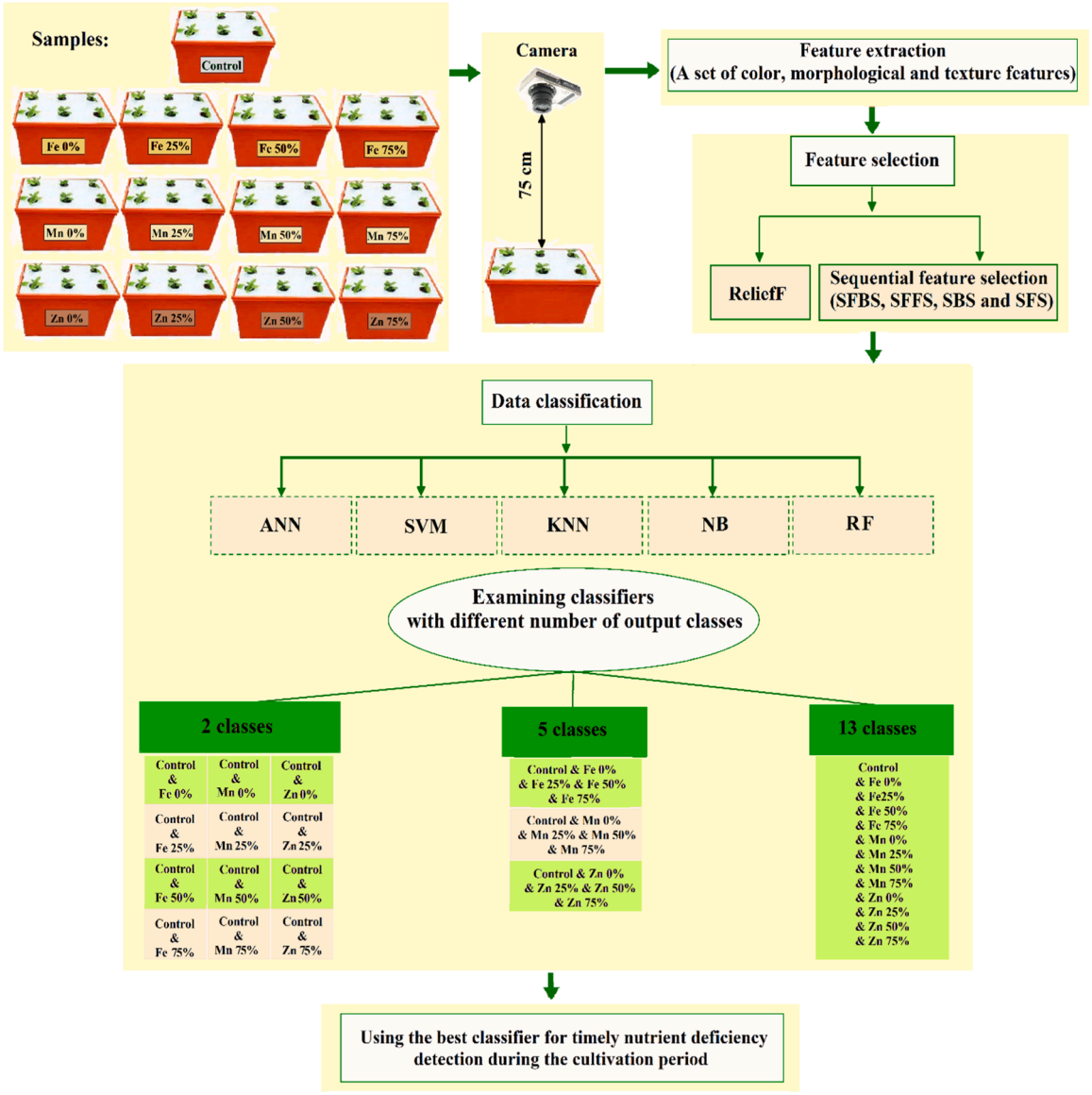


Fig. 1. The stages of the nutrient deficiency detection process in baby spinach applied in this study.

period using a CASIO digital camera (model Exilim EX-ZR700, made in Japan) equipped with a fixed 25 mm equivalent focal length wide-angle lens and an f/3.5 aperture. The camera was positioned above the canopy, perpendicular to the horizontal surface, and maintained at a fixed distance of 75 cm from the canopy of baby spinach plants. Camera parameters were maintained constant throughout the experiment (image resolution: 4608 × 3456 pixels; exposure time: 1/2000 s; ISO 80). The imaging protocol remained unchanged for the entire duration of the experiment. Following the transfer of the images to a computer, image processing operations were performed using the MATLAB 2018b software (The MathWorks, Inc., Natick, MA, USA).

In this study, image processing workflow comprised three stages: preprocessing, segmentation, and feature extraction. During the preprocessing stage, the captured images were enhanced in terms of color using the histogram uniformity method, followed by noise removal through the area opening method. Subsequently, the segmentation process was implemented to divide the images into regions of pixels with similar characteristics. Segmentation based on the homogeneity offers advantages such as reduced storage requirements, faster processing speeds, and ease of modification. Therefore, the plant area was extracted from the image using Otsu's adaptive thresholding method (Fig. 3). In the final step, a comprehensive set of color features (RGB, rgb, Lab, HSV, XYZ, I1I2I3 and h* and Chroma), morphological (area) and texture features obtained from the GLCM (Contrast, Energy, Entropy, Homogeneity, Correlation and Prominence) method were calculated for the final image [18,31]. To determine the symptoms associated with each deficiency level, the neighborhood in the gray level co-occurrence matrix was analyzed in five predefined distances between pixels ($D = 1$ to 5) and in four different directions (0, 45, 90 and 135 degrees).

2.3. Feature selection and classification

Feature selection was performed in accordance with the study by Mallikarjuna and Guru [32], utilizing the ReliefF algorithm, and the Sequential Feature Selection algorithm which had four different modes (Sequential Forward Selection (SFS), Sequential Backward Selection (SBS), Sequential Floating Forward Selection (SFFS) and Sequential Floating Backward Selection (SFBS)). In these algorithms, data extracted from all the images of the plants captured throughout the entire cultivation period were used. It should be noted that the performance of each feature selection method was obtained during 25 iterations, and in each iteration a set of features resulting in the highest accuracy was saved



Fig. 2. Cultivation of baby spinach plants under hydroponic conditions in the greenhouse.

[33].

Subsequently, a set of statistical and intelligent classification methods including ANN, NB, KNN, RF and SVM were used to determine the highest level of accuracy in detecting Fe, Zn and Mn deficiency levels. To compare these five classifiers, the specified features by the most suitable feature selection algorithm were used as input data in a predetermined order, and the classification accuracy for each method with a different number of inputs (1 to 51 features) in several repetitions was obtained. Moreover, the number of different classes were considered for evaluating these classification methods (Table 1). In addition, to evaluate the efficiency of these algorithms, the Monte Carlo Cross Validation method and the k-Fold method were used following the procedures outlined by Xu and Liang [34] and Anguita et al [35], respectively. In fact, because the issue of overfitting is one of the common problems in the field of machine learning algorithms, techniques such as cross validation and early stopping were considered to avoid this problem.

The selection of classification methods (ANN, NB, KNN, RF and SVM) was based on their extensive application in image processing problems, complex data analysis, and their ability to generalize models under different conditions. Moreover, these algorithms were chosen to encompass a broad spectrum of traditional machine learning methods, statistical approaches, and ensemble learning techniques. To properly evaluate classifier performance, the data were randomly split into two sets: 70 % for training and 30 % for testing.

ANN were employed as one of the most powerful nonlinear learning models for plant image classification. In this study, a Multilayer Perceptron (MLP) with a three-layer architecture (input, hidden, and output layers) and the Backpropagation learning method was utilized. The Levenberg-Marquardt training function was selected due to its fast convergence rate and optimal performance in classification tasks [17]. Through comprehensive experimental analyses, it was established that the optimal number of neurons within the hidden layer is six. Further augmentation beyond this threshold did not yield a statistically significant improvement in classification accuracy. The NB algorithm, a probabilistic method based on Bayes' theorem, was implemented in this study following the approach presented by Chou et al [36]. The SVM is a widely used classification technique for high-dimensional data, and in this study, it was implemented according to the method proposed by Yang et al [37]. This model was chosen due to its ability to maximize the separation margin between classes. The KNN algorithm is a non-parametric distance-based method that classifies new samples based on their similarity to training data. In this study, the optimal value of k was determined through multiple experimental trials. This method was implemented following the approach described in Sohail et al [38]. The RF algorithm, an ensemble learning method, enhances model accuracy by combining multiple decision trees while reducing the risk of overfitting. This model was implemented based on the methodology outlined by Ansori et al [39].

As previously mentioned, three different levels of output classification were considered (Table 2) to evaluate the effectiveness of the classifiers across varying levels of complexity. At the first level (binary classification), the primary distinction between healthy and nutrient-deficient plants was evaluated, serving as an initial screening step in plant health monitoring. The second level (five-class classification) aimed to differentiate deficiencies of Fe, Mn, and Zn across four severity levels, alongside a control group, providing valuable insights for plant nutritional management. Finally, the third level (thirteen-class classification) represented the most complex scenario, distinguishing all severity levels of each nutrient to evaluate the models' ability to classify diverse nutrient deficiency conditions.

Subsequently, after identifying the highest-performing classifiers at each stage of assessing different output classes (2, 5, and 13 output classes), their corresponding input features were employed in the best classifier for the timely detection of iron, zinc, and manganese nutrient deficiencies in plant. Then, every six days throughout the cultivation period, the results of expert evaluations of the samples were compared with the outcomes of this classifier.

2.4. Hierarchical model for nutrient deficiency detection

In this section of the study, following the identification of the optimal classification algorithm for detecting nutrient deficiencies, another model, termed the hierarchical model, was developed and implemented to systematically identify stress induced by nutrient deficiency. This model consisted of three consecutive stages that assessed the plant's nutritional status. Fig. 4 illustrates the overall process of the proposed model. As shown in Fig. 4, in the proposed model, after extracting a set of features from the plant image, the classification process was performed hierarchically in three sequential stages: 1) identifying the overall condition of the plant (control or deficient), 2) determining the type of deficiency (Fe, Mn, or Zn) if a deficiency was detected, and 3) assessing the severity of the nutrient deficiency at four different levels (0 %, 25 %, 50 %, 75 %). This model was designed as a hierarchical and multi-stage approach to enable a more precise analysis of the plant's nutritional status while reducing classification complexity.

3. Results and discussion

3.1. Feature selection results

Performance of the utilized feature selection methods is presented in Table 2. The SFBS method had the highest correct classification rate (CCR = 0.556) compared to the other methods used for the classification of the output classes. Following SFBS, the feature selection algorithms were ranked in order of effectiveness as follows: SFFS, SFS, SBS, and ReliefF. Consequently, the results of the SFBS method were utilized in the subsequent classification algorithms.

As shown in Table 2, among the five algorithms compared, SFBS achieved the best performance in terms of correct classification rate. Fig. 5 shows a comparison of CCR values for the different algorithms, clearly demonstrating the superiority of SFBS.

As shown in Fig. 5, the SFS and SBS algorithms, due to their one-directional nature, were unable to escape local minima and consequently resulted in lower CCR values. ReliefF, relying on independent weighting of features, was not effective in capturing complex interactions. SFFS achieved better performance than SFS; however, it still exhibited more limitations compared to SFBS. The floating structure of

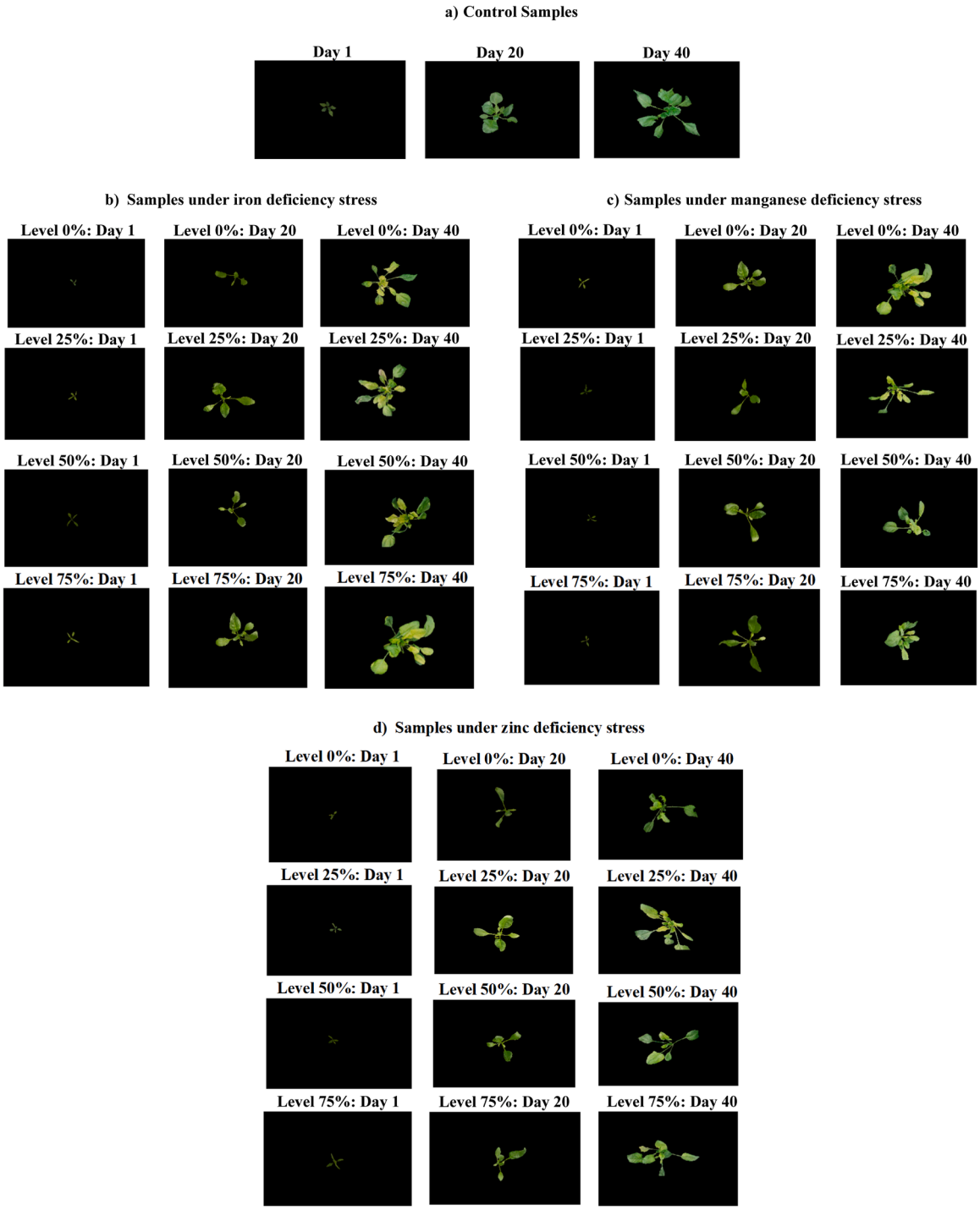


Fig. 3. Implementation of image processing operations to extract plant area from the images of the following treatments: a) control, b) under iron deficiency stress, c) under manganese deficiency stress, and d) under zinc deficiency stress. Day 1 is defined as the day when the first true leaf was expanded.

SFBS, which allows both removal and reintroduction of features, enabled this method to achieve the highest correct classification rate and the most stable lower bound of CCR among all the compared algorithms.

The number of occurrences of each feature after implementing the SFBS algorithm during 25 iterations in Fig. 6. All features were arranged in the order of the most occurrence in different iterations. Finally, these features were used as the input data to the classifiers according to the order shown in the diagram (Fig. 6).

3.2. Performance results of classification algorithms

3.2.1. Binary classification models for distinguishing micronutrient deficiency (Fe, zn, mn) from the control group

It is important to note that the input data of classifiers covered the entire cultivation period (14 days of data collection) in order to assess the performance of the classifiers in detecting nutrient deficiencies. After examining the desired classification methods, the number of input features was determined when each of these classifiers had the highest level of accuracy in detecting the different levels under Fe deficiency stress (Table 3). Furthermore, comparison of the control group with each

Table 1
Output classes of the classification operations.

Number of output classes	Description of the classes
Two	Comparison of the control group with each level of 0 %, 25 %, 50 % and 75 % of iron, manganese and zinc elements
Five	ü Comparison of control group with four levels of iron deficiency ü Comparison of the control group with four levels of manganese deficiency ü Comparison of control group with four levels of zinc deficiency
Thirteen	Comparison of the control group with three nutrients iron, zinc and manganese at four different levels

Table 2
Results of feature selection using Sequential Forward Selection, Sequential Backward Selection, Sequential Floating Forward Selection, Sequential Floating Backward Selection and ReliefF algorithms.

Method of feature selection	Correct classification rate
Sequential Forward Selection	0.506 ± 0.005
Sequential Floating Forward Selection	0.543 ± 0.003
Sequential Backward Selection	0.501 ± 0.003
Sequential Floating Backward Selection	0.556 ± 0.002
ReliefF	0.397 ± 0.006

of the four different levels of Fe deficiency (0 %, 25 %, 50 % and 75 %) can be seen in Fig. 7a. ANN shows the highest level of accuracy in detecting the control group with each of the four levels of Fe deficiency compared to the other classification methods. The number of the most suitable input features in ANN for each of the levels of 0 %, 25 %, 50 % and 75 % were obtained as 6, 10, 6 and 18 features, respectively (Table 3). Following the ANN algorithm, the best classifiers in detecting Fe deficiency at 0 % level compared to the control group are KNN, SVM, NB and RF, respectively. The priority order of the classifiers for separating the control group and the plants under Fe stress of 25 % was also similar to that of 0 %, but with the difference that the NB method was in the third place and the SVM method was in the fourth place. At the 50 % Fe level within the binary classification groups, the ANN method achieved the highest classification accuracy, followed by SVM. The next best-performing classifiers for detecting the 50 % Fe level were NB, KNN, and RF, respectively. However, at the 75 % Fe level, while ANN and SVM remained the top two methods, RF ranked third, and NB had the lowest classification accuracy. The best method in detecting each of the four different levels of Fe deficiency (0 %, 25 %, 50 % and 75 %) compared to the control group are ANN (92.7 %), SVM (86.2 %), KNN (84.0 %), NB (80.5 %) and RF (80.1 %) (Fig. 7a). As shown in Table 4, the Artificial Neural Network (ANN) achieved the highest values of Precision, Recall, and F1-score across all deficiency levels (0 %, 25 %, 50 %, and 75 %) compared to the other classifiers. The mean values of the three parameters (Precision = 93.96 %, Recall = 92.50 %, F1-score = 92.53 %) indicate the good performance of this method. Following ANN, the Support Vector Machine (SVM) ranked second with relatively high mean values (Precision = 87.85 %, Recall = 85.95 %, F1-score = 86.08 %). In the subsequent ranks, KNN (F1-score = 83.60 %), RF (F1-score = 79.84 %), and NB (F1-score = 79.76 %) were placed, showing a considerable reduction in Recall and F1-score, particularly at higher deficiency levels (50 % and 75 %).

The priority order of the classification methods for each stress level of Mn deficiency are: ANN, RF, KNN, NB and SVM at the 0 % level; ANN, RF, NB, SVM and KNN at the 25 % level; ANN, Random Forest, KNN, SVM and NB at 50 % level; and ANN, RF, SVM, KNN and NB at the 75 % level (Fig. 7b). The ANN algorithm is also highly accurate in detecting each of the four levels of Mn nutrient deficiency compared to the control group (Fig. 7b), and the number of input features of this method for each level of 0 %, 25 %, 50 % and 75 % was 13, 11, 6 and 3, respectively (Table 3). The RF method has the highest detection accuracy (83.3 %) after ANN. Following those, the KNN (78.6 %), SVM (75.3 %) and NB (71.6 %) algorithms were the best classification methods, respectively (Fig. 7b). The results of Precision, Recall, and F1-score for the binary classification of manganese deficiency are presented in Table 4. As

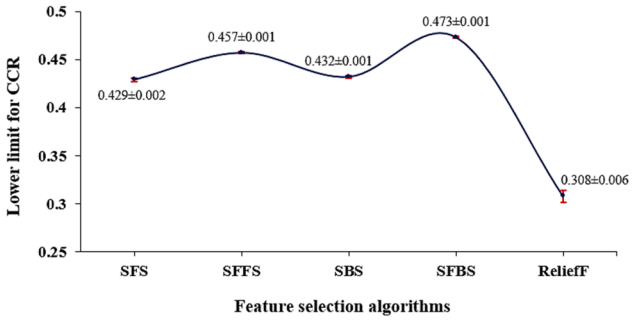


Fig. 5. Comparison of lower limit results for CCR between different feature selection algorithms.

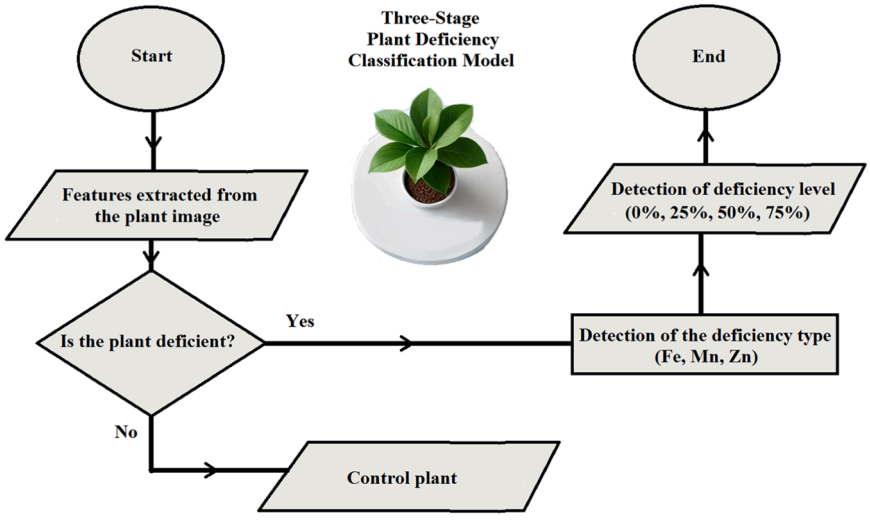


Fig. 4. Flowchart of the proposed hierarchical model.

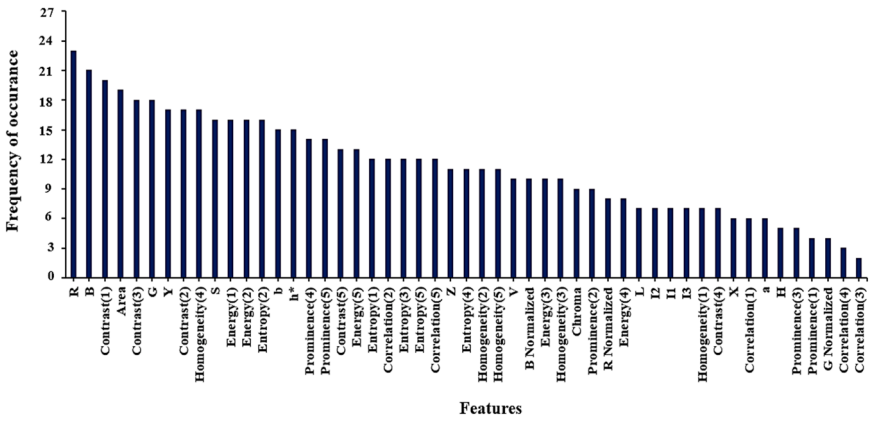


Fig. 6. Ranked feature selection results based on occurrence frequency using the Sequential Floating Backward Selection algorithm.

Table 3
The number of input features of five classification methods with two output classes.

Stress levels of the nutrient elements		Classification algorithms				
		Artificial Neural Network	Support Vector Machine	K-Nearest Neighbors	Random Forest	Naive Bayes
Iron	Level 0%	6	2	2	4	16
	Level 25%	10	2	2	7	14
	Level 50%	6	2	2	9	38
	Level 75%	18	2	11	3	16
Manganese	Level 0%	13	2	2	7	12
	Level 25%	11	2	2	6	47
	Level 50%	6	2	2	8	7
	Level 75%	3	2	2	19	16
Zinc	Level 0%	6	2	2	10	8
	Level 25%	5	2	2	2	6
	Level 50%	6	2	22	6	34
	Level 75%	10	7	20	37	40

Algorithms that had the highest classification accuracy are highlighted in gray.

shown, ANN obtained the highest mean values (Precision = 88.65 %, Recall = 87.18 %, F1-score = 87.14 %) and consistently outperformed the other classifiers across all deficiency levels (0 %, 25 %, 50 %, and 75 %). RF ranked second, with relatively close mean values (Precision = 84.61 %, Recall = 82.83 %, F1-score = 82.71 %), demonstrating good performance at some deficiency levels (notably at 50 %). In the subsequent ranks, KNN (F1-score = 78.03 %), SVM (F1-score = 74.85 %), and NB (F1-score = 71.03 %) showed lower values, with a marked decrease in Recall and F1-score, particularly at the 75 % deficiency level.

For Zn nutrient deficiency, based on the average values of recognition accuracy of the desired classifiers, ANN (88.3 %), KNN (82.7 %), SVM (78.4 %), RF (78.1 %) and NB (73.3 %) were reported as the most suitable classification methods in identifying each of the four stress levels of 0 %, 25 %, 50 % and 75 % of deficiency compared to the control group (Fig. 7c). The number of input features in the ANN algorithm for each of the 0 %, 25 %, 50 % and 75 % levels was 6, 5, 6 and 10 features, respectively (Table 3). The order of preference of the classification method for each stress level of Zn deficiency was ANN, SVM, KNN, RF and NB at 0 % level; ANN, KNN, NB SVM and RF at 25 % level; ANN, SVM, KNN, RF and NB at the 50 % level; and ANN, KNN, RF, SVM and NB at the 75 % level (Fig. 7c). According to the Table 4, in the evaluation of classifiers for detecting zinc deficiency levels, ANN achieved the best performance, with the highest mean values of Precision = 90.10 %, Recall = 88.04 %, and F1-score = 87.86 %. KNN ranked second, with acceptable mean values (Precision = 84.80 %, Recall = 82.25 %, F1-score = 82.35 %), providing a relatively better balance among the three metrics compared to other classifiers. SVM and RF showed comparable results, with F1-scores of 77.88 % and 77.42 %, respectively, ranking third and fourth. However, both methods showed a sharp decline in Recall and F1-score at the 75 % deficiency level, indicating lower sensitivity to samples with higher stress intensity. Finally, NB exhibited the weakest performance, with the lowest mean values (Precision = 75.28 %, Recall = 72.75 %, F1-score = 72.37 %).

To provide an overall assessment of the performance of classification models in binary classification, the average classification accuracy of each method was calculated across binary classification scenarios. In these scenarios, each stress level of nutrient deficiencies (Fe, Zn, and Mn) was independently distinguished from the control group. Accordingly, the comparative analysis of the five classification methods revealed that ANN demonstrated the highest effectiveness with an accuracy of 89.5 %, followed by KNN (81.8 %), RF (80.4 %), SVM (80.0 %), and NB (75.1 %). In addition, ANN achieved the best overall performance with mean values of Precision = 90.90 %, Recall = 89.24 %, and F1-score = 89.18 %. Following ANN, KNN (Precision = 84.14 %, Recall = 81.29 %, F1-score = 81.33 %) and RF (Precision = 83.33 %, Recall = 80.13 %, F1-score = 79.99 %) showed acceptable performance. In contrast, SVM (Precision = 81.90 %, Recall = 79.65 %, F1-score = 79.60 %) and NB (Precision = 77.27 %, Recall = 74.75 %, F1-score = 74.39 %) exhibited weaker results compared to the other methods. This indicates that ANN maintained its superiority not only in terms of accuracy but also across the other evaluation parameters.

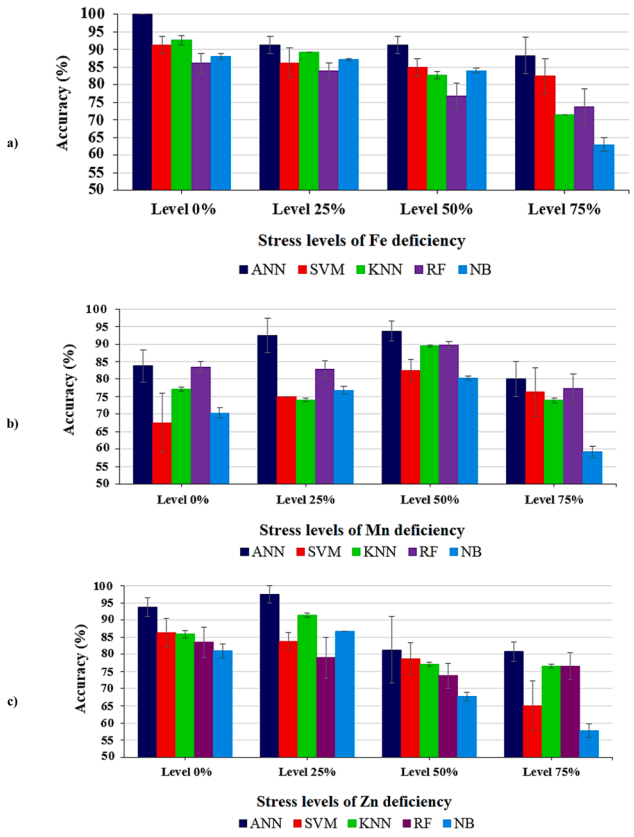


Fig. 7. Comparison of the performance of several different algorithms for the classification of the control group and each of the four different levels of nutrient deficiency: (a) Iron, (b) Manganese, and (c) Zinc.

3.2.2. *Five-class classification models for differentiating micronutrient stress levels (Fe, zn, mn) in 25 % intervals*

The results of the different classification methods with five output classes (control group and four levels of deficiency of a nutrient element) for the three nutrients of Fe, Zn and Mn elements are shown in Fig. 8. The best classification method was the ANN algorithm with an average detection accuracy of 47.8 %. The number of input features of this classifier in the groups under the stress of Fe, Mn and Zn deficiency was 7, 24 and 28, respectively (Table 5). After this method, the ranking of classification methods was: KNN (40.1 %), RF (30.4 %), NB (30.0 %) and SVM (29.3 %). According to Table 6, the results of Precision, Recall, and F1-score confirm that ANN still achieved the highest performance, with mean values of Precision = 47.50 %, Recall = 47.66 %, and F1-score = 45.51 %. KNN ranked second, with Precision = 40.77 %, Recall = 40.33 %, and F1-score = 38.29 %. The other methods demonstrated lower performance: NB (Precision = 27.83 %, Recall = 30.00 %, F1-score = 27.23 %), RF (Precision = 23.65 %, Recall = 31.16 %, F1-score = 25.42 %), and SVM (Precision = 26.01 %, Recall = 28.66 %, F1-score = 22.93 %).

3.2.3. *Thirteen- class classification models for simultaneously distinguishing different deficiency levels of Fe, Zn, and Mn*

To evaluate the effectiveness of different classification methods in identifying varying levels of nutrient deficiencies, a thirteen-class classification framework was implemented. After examining the desired classification methods with the number of thirteen output classes (control group and all levels of Fe, Zn and Mn nutrient deficiencies), the ANN with the recognition accuracy of 17.0 % and the number of 8 input features (Table 7) had the best performance, followed by KNN (16.0 %), RF (11.4 %), SVM (11.2 %) and NB (11.0 %) (Fig. 9). In the thirteen-class scenario, ANN achieved mean values of Precision = 15.96 %, Recall = 17.50 %, and F1-score = 14.68 %. KNN showed relatively higher values than ANN, with Precision = 23.67 %, Recall = 16.92 %, and F1-score = 15.82 %. The other classifiers demonstrated weaker performance: RF (Precision = 14.35 %, Recall = 12.50 %, F1-score = 10.31 %), NB (Precision = 13.06 %, Recall = 11.15 %, F1-score = 10.35 %), and SVM (Precision = 8.34 %, Recall = 10.57 %, F1-score = 6.81 %). These findings indicate that under more complex multiclass conditions, KNN and ANN outperform the other methods.

By averaging the classification accuracies of each method across different numbers of output classes, a ranking of the classification methods was established. Based on these classification results, it can be stated that the order of superiority of the desired methods in determining different nutrient stress levels for Fe, Zn, and Mn are ANN (51.43 %), KNN (45.96 %), RF (40.73 %), SVM (40.16 %), and NB (38.70 %). It is noteworthy that these five different classification algorithms showed the highest performance, with an average detection accuracy of 81.3 %, when the number of stress levels was limited to one. However, the performance of all these methods in accurately detecting the status of samples with four levels of stress dropped to below 50 %, with average

Table 4
Precision, Recall, and F1-score parameters of several different algorithms in binary classification (control group and each of the four different levels of nutrient deficiency of Iron, Manganese, and Zinc).

Stress levels of the nutrient elements		Classification algorithms														
		Artificial Neural Network			Support Vector Machine			K-Nearest Neighbors			Random Forest			Naive Bayes		
		Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)
Iron	0%	100	100	100	92.80	91	90.86	93.60	92	91.86	89.94	86.33	85.94	87.77	87.50	87.30
	25%	92.80	91	90.86	88.25	86	86.40	90.90	89	89.15	85.66	84	84.26	90.21	87	86.21
	50%	92.80	91	90.86	87.25	85	85.40	83.40	82	81.90	81.16	76	75.66	88.11	84	83.64
	75%	90.25	88	88.40	83.13	81.80	81.66	72	71	71.50	74	73	73.50	63.31	62.50	61.90
	Mean	**93.96**	**92.50**	**92.53**	**87.85**	**85.95**	**86.08**	**84.97**	**83.50**	**83.60**	**82.69**	**79.83**	**79.84**	**82.35**	**80.25**	**79.76**
Manganese	0%	84.60	83	82.86	70.52	67	66.01	82.16	77	76.66	84.44	83.33	83.06	75.01	70	69.41
	25%	93.60	92	91.86	75	75	75	79.16	74	73.66	83.40	82	81.90	81.16	76	75.66
	50%	95	93.75	93.65	83.40	82	81.90	91.15	88.50	89.15	89.33	89	89.16	80.74	80	80.03
	75%	81.43	80	80.20	77	76	76.50	78.16	73	72.66	81.27	77	76.71	59.77	59	59.03
	Mean	**88.65**	**87.18**	**87.14**	**76.48**	**75**	**74.85**	**82.66**	**78.12**	**78.03**	**84.61**	**82.83**	**82.71**	**74.17**	**71.25**	**71.03**
Zinc	0%	95	93.75	93.65	88.25	86	86.40	87.25	85	85.40	87.27	83	82.71	82.33	81	81.10
	25%	97.43	97	97.23	87.61	83	82.27	92.80	91	90.86	84.16	79	78.66	88.25	86	86.40
	50%	86.66	81.25	80.31	83.16	78	77.66	78	77	77.50	78.16	73	72.66	70.54	67	65.66
	75%	81.33	80.16	80.26	66.41	65	65.20	81.16	76	75.66	81.16	76	75.66	60	57	56.34
	Mean	**90.10**	**88.04**	**87.86**	**81.36**	**78**	**77.88**	**84.80**	**82.25**	**82.35**	**82.69**	**77.75**	**77.42**	**75.28**	**72.75**	**72.37**

Algorithms that had the highest classification accuracy are highlighted in gray.
Pre = Precision; *Re* = Recall; F1 = F1-score.

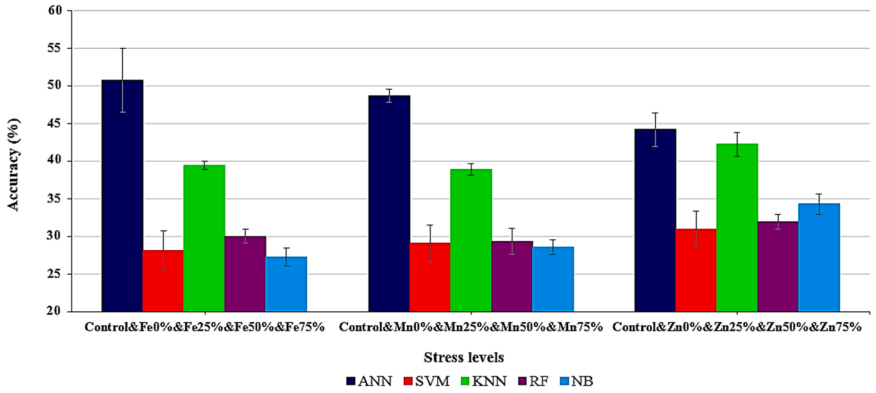


Fig. 8. The results of several different classification methods with five output classes.

Table 5
The number of input features of five different classification methods with five output classes.

Output classes	Classification algorithms				
	Artificial Neural Network	Support Vector Machine	K-Nearest Neighbors	Random Forest	Naive Bayes
Control & Fe0% & Fe25% & Fe50% & Fe75%	7	6	2	11	11
Control & Mn0% & Mn25% & Mn50% & Mn75%	24	26	2	8	44
Control & Zn0% & Zn25% & Zn50% & Zn75%	28	2	2	8	48

Algorithms that had the highest classification accuracy are highlighted in gray.

Table 6
Precision, Recall, and F1-score parameters of several different algorithms in five-class classification (control group and each of the four different levels of nutrient deficiency of Iron, Manganese, and Zinc).

Stress levels of the nutrient elements	Classification algorithms														
	Artificial Neural Network			Support Vector Machine			K-Nearest Neighbors			Random Forest			Naive Bayes		
	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)	Pre (%)	Re (%)	F1 (%)
Iron	47.71	49.50	46.57	29.48	30	21.41	37.38	39	36.17	27.50	32.50	28.10	32.33	30	28.44
Manganese	50.36	47	45.25	17.60	25.50	18.25	33.23	38	34.62	24.28	28.50	24.62	24.16	28	24.60
Zinc	44.42	46.50	44.72	30.95	30.50	29.14	51.71	44	44.09	19.19	32.50	23.53	27	32	28.66
Mean	47.50	47.66	45.51	26.01	28.66	22.93	40.77	40.33	38.29	23.65	31.16	25.42	27.83	30	27.23

Algorithms that had the highest classification accuracy are highlighted in gray.
Pre = Precision; *Re* = Recall; F1 = F1-score.

Table 7
The number of input features of five different classification methods with thirteen output classes.

Classification algorithms	Accuracy (%)	Number of features
Artificial Neural Network	17.0	8
Support Vector Machine	11.2	33
K-Nearest Neighbors	16.0	2
Random Forest	11.4	20
Naive Bayes	11.0	51

Algorithms that had the highest classification accuracy are highlighted in gray.

detection accuracy of 35.5 %. When all the levels of Fe, Zn and Mn nutrients were considered as the output of these classifiers, the performance of all methods significantly decreased to below 20 %, with an average detection accuracy of 13.3 %. Beyond overall accuracy, the comparison of mean Precision, Recall, and F1-score across all classification scenarios (binary, five-class, and thirteen-class) showed that ANN achieved higher values than the other methods (Precision = 67.24 %, Recall = 66.57 %, and F1-score = 65.54 %). KNN ranked second (Precision = 63.37 %, Recall = 60.22 %, F1-score = 59.62 %), followed by RF (Precision = 57.60 %, Recall = 56.81 %, F1-score = 55.14 %). SVM (Precision = 56.01 %, Recall = 55.63 %, F1-score = 53.71 %) and NB (Precision = 54.54 %, Recall = 53.08 %, F1-score = 52.15 %) showed lower results. These findings confirm that the superiority of ANN was not limited to overall accuracy but was also evident in the other

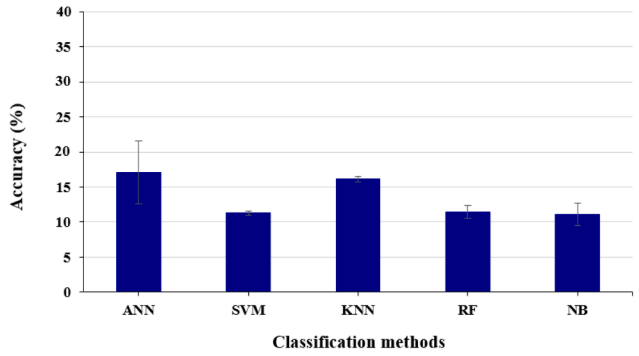


Fig. 9. The results of several different classification methods with thirteen output classes.

evaluation parameters.

In another study, Rahadiyan et al [40] classified chili pepper plants based on calcium, magnesium, sulfur, and multiple deficiencies using RGB, GLCM, Hu moments, and Distance of Centroid features. CNN achieved the highest accuracy (97.76 %), followed by SVM (90.55 %) and MLP (89.70 %). Nadafzadeh et al [17] monitored Fe deficiency in spinach plants, achieving a classification accuracy of 96 % using ANN. Additionally, Fe content prediction was performed with an SVR model ($R^2 = 0.94$), demonstrating the potential of machine learning in nutrient quantification. Kavitha [41] applied image processing techniques to identify nitrogen, phosphorus, and potassium deficiencies in rice plants. Using RF and NB, the highest classification accuracy was obtained with RF (94.66 % for nitrogen, 89.63 % for phosphorus, and 89.71 % for potassium), while NB showed comparatively lower performance. Rodrigo [42] employed CNN-based YOLOv3 architecture for detecting calcium, nitrogen, and magnesium deficiencies in greenhouse lettuce. The model achieved an accuracy of 94.38 % in the training phase and 75.53 % in the test phase.

Among the evaluated classifiers in the present study, the ANN exhibited the highest overall performance, demonstrating its ability to effectively recognize subtle variations and intricate patterns associated with different plant stress conditions. This confirms the advantage of ANN in handling non-linear relationships within agricultural datasets, making it a suitable choice for nutrient deficiency classification. KNN, identified as the second most effective method, proved to be a reliable and easy-to-implement algorithm. However, its efficiency is influenced by the choice of K, as larger values slow down the prediction process. Since the RF algorithm ranked third in performance, it demonstrated a moderately effective classification approach. Despite its advantages in feature selection and overfitting reduction, RF's reliance on multiple decision trees increases processing time and computational demands, which may limit its practicality in real-time agricultural applications. SVM demonstrated moderate performance, performing well in high-dimensional feature spaces but showing limitations when dealing with overlapping class distributions. Meanwhile, the NB algorithm, despite being computationally efficient and requiring minimal training data, exhibited the lowest classification accuracy due to its assumption of feature independence, which does not always align with real-world agricultural scenarios.

Furthermore, the results of this study revealed that while all models performed well in distinguishing a smaller number of stress levels, their accuracy declined as classification complexity increased. This underscores the importance of selecting classification methods based on the complexity of the task. Unlike previous studies that primarily focused on individual nutrient deficiencies or binary classifications, this research provided a broader evaluation across multiple stress levels.

3.3. Early detection of nutrient deficiencies

In accordance with the results of Section 3.2, since, in each classification scenario, only one of the five classifiers with specific input features achieved the highest performance (Tables 3, 5, and 7), all the input data from these top-performing classifiers were re-employed in the ANN, which was identified as the most effective classifier among the studied models. The total of these features are the first 28 features in Fig. 6, which according to the results of the SFBS algorithm had the highest frequency of occurrence (R, B, Contrast(1), Area, Contrast(3), G, Y, Contrast(2), Homogeneity(4), S, Energy(1), Energy(2), Entropy(2), b, h*, Prominence(4), Prominence(5), Contrast(5), Energy(5), Entropy(1), Correlation(2), Entropy(3), Entropy(5), Correlation(5), Z, Entropy(4), Homogeneity(2), Homogeneity(5)). Fig. 10 shows the results of the evaluation of the samples by the expert and the classifier during the cultivation period. Notably, the objective of this approach was to facilitate the timely detection of Fe, Zn, and Mn in plants.

According to Fig. 10a to Fig. 10c, at the onset of the cultivation period, when the true leaves of the plant emerged, deficiencies in the three nutrients Fe, Mn, and Zn could be detected with accuracies of 88 %, 84 % and 76 %, respectively. Despite observed fluctuations, the detection accuracy gradually decreased until the midpoint of the plant cultivation period. But, as the plants aged and surpassed the halfway of the cultivation period, more pronounced stress symptoms appeared, leading to an increase in diagnostic accuracy. Notably, it was challenging to visually to recognize the stress in the early days of the growth period, with visible signs of stress appearing only after two weeks. Throughout the data collection period, the classifier's accuracy in identifying the deficiency levels of these three nutrients consistently surpassed the accuracy of visual recognition by humans. It should be mentioned that the both performance of the classifier and human visual ability to distinguish all stress levels of the three nutrients Fe, Mn and Zn during the cultivation period was very weak, with accuracies below 50 % (Fig. 10d). As shown in Fig. 10d, when the plant age exceeded the midpoint of the cultivation period, human visual identification of these nutrient deficiencies was more accurate than the classifier.

In a related study, a system was developed that leveraged entropy, energy, and homogeneity features extracted from cucumber plant images. Using a regression-based algorithm, nitrogen deficiency was detected approximately two days before human observation [43]. Similarly, Xu et al [44] employed an artificial neural network to distinguish nitrogen and potassium deficiencies in tomato plants with an accuracy of 82.5 %, identifying stress conditions 6 to 10 days earlier than human assessment. Furthermore, Story et al [25] utilized a regression approach to detect calcium deficiency in lettuce plants one day before visible symptoms appeared. These studies collectively highlight the capability of computational techniques to identify nutrient deficiencies before they become visually apparent. The findings of the present study align with this body of research, further demonstrating the advantages of machine learning models for early nutrient deficiency detection. More specifically, this study reinforces the efficacy of ANN in detecting iron, zinc, and manganese deficiencies at early growth stages. This study examined the progression of deficiencies throughout the entire cultivation period. The results indicate that the classifier consistently maintained high accuracy from the early stages of plant growth, whereas human visual assessment improved only as stress symptoms became more pronounced in later stages. These findings underscore the potential of machine learning models not only for early detection but also for minimizing reliance on subjective human evaluations, ultimately contributing to more precise and efficient agricultural practices.

3.4. Results of a hierarchical model assessment for monitoring plant nutritional status

Since plant nutrient deficiencies exhibit a hierarchical structure (i.e., first, it must be determined whether the plant is deficient, then the type

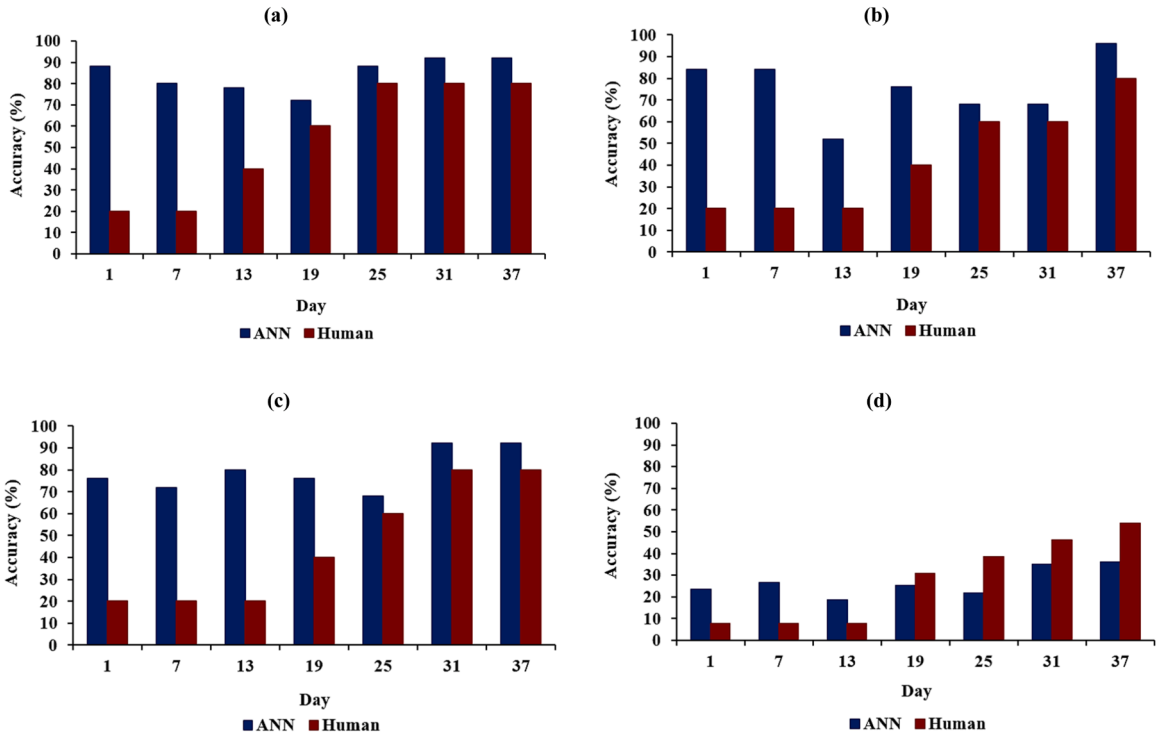


Fig. 10. The performance results of the classifier and the expert person to detect: a) four stress levels of iron deficiency, b) four stress levels of manganese deficiency, c) four stress levels of zinc deficiency, and d) all the stress levels of the three nutrients iron, manganese and zinc. Day 1 is defined as the day when the first true leaf was expanded.

of deficiency must be identified, and finally, its severity assessed), the adoption of a three-stage model based on the most suitable classification algorithm in this domain (ANN) was considered. Although Section (3.2) demonstrated that ANN outperform other classification methods in detecting nutrient deficiencies, the use of the three-stage hierarchical model was considered for two main reasons: 1) to reduce model complexity by decomposing the classification process into three distinct stages, and 2) to minimize errors caused by data overlap by analyzing a limited subset of data at each stage.

Ultimately, by utilizing a set of features (as identified in Section 3.3) as input data for the proposed model, an overall accuracy of 72.97 % ± 0.54 was achieved. In addition, the mean values of Precision, Recall, and F1-score were 69.34 %, 70.70 %, and 68.34 %, respectively. This result indicates that the proposed model demonstrates a relatively reasonable capability in differentiating data associated with nutrient deficiencies. However, the evaluations suggest that while the decision-making structure has become clearer, the overlap of features among different deficiency types still leads to a reduction in model accuracy. Specifically, determining the severity of the deficiency (in the third stage of the model) proved to be more challenging than detecting the general presence of deficiency (in the first stage). This suggests that the extracted dataset may not provide sufficient resolution for such detailed differentiation. Consequently, since spectral imaging facilitates the extraction of a larger volume of data, it is recommended that future research investigate this method to enhance the model's performance.

4. Conclusion

Timely identification of nutrient deficiency is essential for maintaining plant health and maximizing productivity, particularly in controlled cultivation systems such as hydroponics. This study explored the detection of Fe, Zn, and Mn deficiencies in baby spinach using a combination of statistical and machine learning classification techniques. Among the evaluated models, artificial neural networks demonstrated the highest efficacy, highlighting their potential as a powerful tool for monitoring plant nutritional status. The findings emphasize the challenges associated with classifying multiple deficiency levels, as detection accuracy declines with increasing number of output classes. Despite these challenges, the hierarchical classification approach exhibited promising performance, suggesting its applicability for real-time nutrient deficiency monitoring. This study highlights the potential of computational methods in modern agriculture, offering a pathway toward more precise and efficient nutrient management practices. By identifying deficiencies before visible symptoms appear, such technologies can play a vital role in minimizing crop stress and optimizing resource allocation. This study was conducted in a controlled greenhouse environment and based on a single dataset to determine the independent effects of nutrient deficiencies. While this approach has the advantage of eliminating unwanted factors (such as fluctuations in light, temperature, humidity, and soil characteristics) and increasing the accuracy of analysis, it also creates limitations for generalizing the results to field conditions and diverse environments. It is therefore suggested that future research utilize multi-environment datasets and different varieties to better evaluate the performance and stability of the model in practical applications and different cultivation conditions.

Ethical statement

In this paper there is no Ethical Statement for experimentation with human and animal subjects

CRediT authorship contribution statement

Maryam Nadafzadeh: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Ahmad Banakar:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Saman Abdanan Mehdizadeh:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Formal

analysis, Data curation. **Saeid Minaei:** Writing – review & editing, Visualization, Validation, Supervision, Formal analysis. **Abdul Mounem Mouazen:** Writing – review & editing, Validation, Software. **Gerrit Hoogenboom:** Writing – review & editing, Visualization, Validation, Supervision.

Declaration of competing interest

There is no declaration of interest for this paper.

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Data availability

Data will be made available on request.

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